⚗️CHEMICAL REACTIONS AND EQUATIONS⚗️

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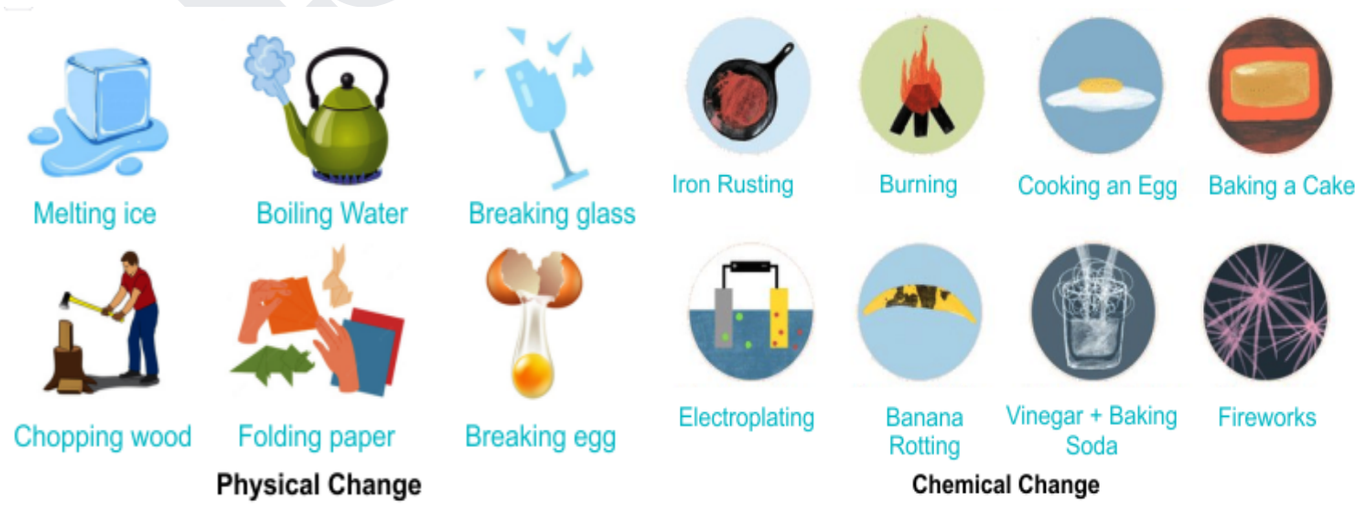
1. Physical vs. Chemical Change

🔵 Physical Change

- Only the state or form of matter changes; composition remains the same.
- The change is often reversible.
- Example: Ice → Water → Ice (by temperature change).

🔴 Chemical Change

- The composition of matter changes permanently.
- New substances with entirely different properties are formed.
- Example: Burning of coal, ripening of fruit, curd formation from milk.



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📊 Comparison Table

Feature	Physical Change	Chemical Change
Composition	Unchanged	Changes permanently
New substance formed	No	Yes
Reversibility	Usually reversible	Usually irreversible
Driving force	Temperature, Pressure	Breaking & forming of bonds
Example	Evaporation of water	Cooking of food

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2. Chemical Reactions & Equations

🔬 What is a Chemical Reaction?

A chemical reaction is a process in which substances undergo bond breaking and are transformed into new substances by formation of new bonds.

- Substances taking part in a reaction $\rightarrow$ **Reactants**
- Substances formed as a result $\rightarrow$ **Products**

Example: Combustion of coal in air

$$\text{Carbon (Coal)} + \text{Oxygen} \rightarrow \text{Carbon dioxide}$$

Here, C and O_2 are reactants; CO_2 is the product.

🧾 Word Equation $\rightarrow$ Chemical Equation

Step 1 — Word Equation:

$$\text{Copper sulphate (aq)} + \text{Zinc dust} \rightarrow \text{Zinc sulphate (aq)} + \text{Copper}$$

Step 2 — Chemical Equation (using formulae):

$$CuSO_4 + Zn \rightarrow ZnSO_4 + Cu \quad \cdots (2)$$

- The ionic bond in $CuSO_4$ breaks; a new ionic bond in $ZnSO_4$ forms.
- Products ($ZnSO_4$, Cu) have totally different properties from the reactants.

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3. Writing a Chemical Equation — Conventions

📋 Rules (with Examples)

Rule 1 — Position of Reactants & Products:

- Reactants $\rightarrow$ Left-hand side (LHS)
- Products $\rightarrow$ Right-hand side (RHS)
- An arrow ($\rightarrow$) in between indicates the direction of the reaction.

Rule 2 — Multiple Reactants/Products:

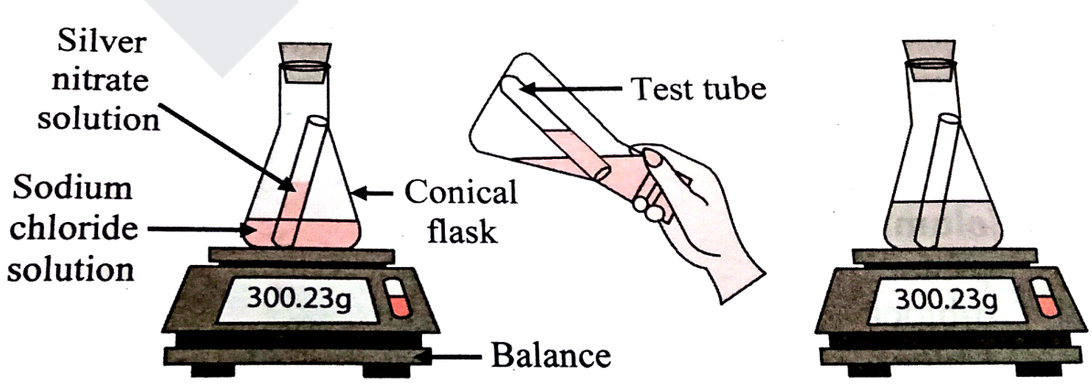
- Linked with a plus sign ($+$).

$$CuSO_4 + Zn \rightarrow ZnSO_4 + Cu$$

Rule 3 — Physical States:

Symbol	State
(g)	Gaseous
(l)	Liquid
(s)	Solid
(aq)	Aqueous solution
$\uparrow$	Gaseous product (evolved)
$\downarrow$	Insoluble solid precipitate

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s) \quad \cdots (3)$$



$$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl\downarrow + NaNO_3(aq) \quad \cdots (9)$$

Rule 4 — Heat Applied from Outside:

- Indicated by the symbol Δ above the arrow.

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2 \uparrow \quad \cdots (4)$$

- If heat is released during the reaction, it is written as a product:

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s) + \text{Heat} \quad \cdots (5)$$

Rule 5 — Special Conditions (Temperature, Pressure, Catalyst):

- Written above or below the arrow.

$$\text{Vegetable oil}(l) + H_2(g) \xrightarrow[\text{Ni Catalyst}]{60^\circ \text{C}} \text{Vanaspathi ghee}(s) \quad \cdots (6)$$

Rule 6 — Special Information about Reactants/Products:

- Written below the respective formula.

$$Cu(s) + \underset{\text{Concentrated}}{4HNO_3(aq)} \rightarrow Cu(NO_3)_2(aq) + 2NO_2(g) + 2H_2O(l) \quad \cdots (7)$$

$$3Cu(s) + \underset{\text{Dilute}}{8HNO_3(aq)} \rightarrow 3Cu(NO_3)_2(aq) + 2NO(g) + 4H_2O(l) \quad \cdots (8)$$

💡 **Note:** NO_2 (reddish, poisonous) is formed with concentrated HNO_3; NO is formed with dilute HNO_3.

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4. Balancing a Chemical Equation

⚖️ Law of Conservation of Mass

In any chemical reaction, the total mass of each element in the reactants equals the total mass of that element in the products.

- An equation where the number of atoms of each element is equal on both sides → **Balanced Equation**
- If unequal → **Unbalanced Equation**

🪜 Step-by-Step Balancing (Trial & Error Method)

Example: $NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$

STEP I — Write the unbalanced equation:

$$NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O \quad \cdots (10)$$

STEP II — Count atoms on each side:

Element	LHS (Reactants)	RHS (Products)	Balanced?
Na	1	2	❌
O	5	5	✅
H	3	2	❌
S	1	1	✅

The equation is unbalanced.

STEP III — Balance by applying factors:

(i) Balance Na: Na_2SO_4 has 2 Na atoms in products; apply factor 2 to $NaOH$:

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O \quad \cdots (10)'$$

⚠️ **Important Rule:** The formula of a compound cannot be changed while balancing. Never write Na_2OH; instead multiply $NaOH$ by 2.

(ii) Balance H: LHS has 4 H atoms; RHS has only 2. Apply factor 2 to H_2O:

H Atoms	LHS ($2NaOH+H_2SO_4$)	RHS (H_2O)
Before	4	2
After factor	4	$2 \times 2 = 4$ ✅

$$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O \quad \cdots (10)''$$

STEP IV — Verify and write the final balanced equation:

Element	LHS	RHS	Balanced?
Na	2	2	✅
O	6	6	✅
H	4	4	✅
S	1	1	✅

$$\boxed{2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O} \quad \cdots (11)$$

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5. Types of Chemical Reactions

Chemical reactions are classified into four types based on the nature and number of reactants and products.

🔗 5.1 Combination Reaction

When two or more reactants combine to form a single product, it is called a Combination Reaction.

- **Key Feature:** Multiple reactants $\rightarrow$ One product

Examples:

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s) \quad \cdots (12)$$

(Ammonia + Hydrogen chloride $\rightarrow$ Ammonium chloride — white smoke)

$$2Mg(s) + O_2(g) \rightarrow 2MgO(s) \quad \cdots (13)$$

(Burning magnesium in air $\rightarrow$ white powder of magnesium oxide)

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + \text{Heat} \quad \cdots (14)$$

(Slaked lime + water $\rightarrow$ calcium hydroxide with large heat evolution)

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💥 5.2 Decomposition Reaction

A reaction in which a single reactant breaks down into two or more products is called a Decomposition Reaction.

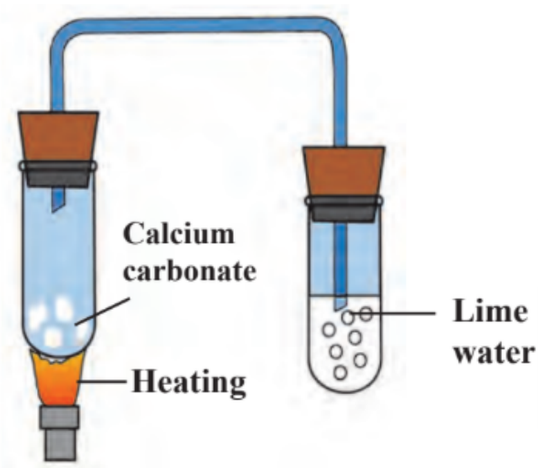
- **Key Feature:** One reactant $\rightarrow$ Multiple products

Examples:

$$C_{12}H_{22}O_{11} \xrightarrow{\text{Heat}} 12C + 11H_2O \quad \cdots(15)$$

(Sugar decomposes into carbon and water on heating)

$$CaCO_3(s) \xrightarrow{\Delta} CaO(s) + CO_2 \uparrow \quad \cdots(16)$$



Decomposition of calcium carbonate

(Calcium carbonate decomposes into calcium oxide + CO_2 ; turns lime water milky)

$$2H_2O_2(l) \rightarrow 2H_2O(l) + O_2 \uparrow \quad \cdots(17)$$

(Hydrogen peroxide slowly decomposes at room temperature)

$$2H_2O(l) \xrightarrow{\text{Electrical Energy}} 2H_2 \uparrow + O_2 \uparrow \quad \cdots(18)$$

(Electrolysis of acidulated water — decomposition by electrical energy)

🔄 5.3 Displacement Reaction

A reaction in which a more reactive element displaces a less reactive element from its compound (by forming its own ions) is called a Displacement Reaction.

- **Key Feature:** Element A replaces Element B in a compound.

Example:

$$CuSO_4(aq) + Zn(s) \rightarrow ZnSO_4(aq) + Cu(s) \quad \cdots(3)$$

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- Zn^{2+} ions replace Cu^{2+} ions in copper sulphate.
- Zinc is more reactive than copper.

Displacement	Equation
Fe displaces Cu	$CuSO_4(aq) + Fe(s) \rightarrow FeSO_4(aq) + Cu$
Pb displaces Cu	$CuSO_4(aq) + Pb(s) \rightarrow PbSO_4(aq) +$

🔀 5.4 Double Displacement Reaction

A reaction in which the ions in the reactants are exchanged to form a precipitate (insoluble product) is called a Double Displacement Reaction.

- **Key Feature:** Two compounds exchange their ions; a precipitate forms.

Example:

$$AgNO_3(aq) + NaCl(aq) \rightarrow \underset{\text{White ppt.}}{AgCl\downarrow} + NaNO_3(aq) \quad \cdots(9)$$

Ag^+ and Na^+ ions exchange places.

📊 Summary Table: Types of Reactions

Type	No. of Reactants	No. of Products	Key Feature	Example
Combination	2 or more	1	$A + B \rightarrow AB$	$2Mg + O_2 \rightarrow 2MgO$
Decomposition	1	2 or more	$AB \rightarrow A + B$	$CaCO_3 \xrightarrow{\Delta} CaCO + CO_2$
Displacement	2	2	More reactive element displaces less reactive	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
Double Displacement	2	2 (+precipitate)	Exchange of ions; ppt. formed	$AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

6. Endothermic & Exothermic Processes and Reactions

🌡️ **Definitions**

- **Endothermic Process/Reaction:** Heat is absorbed from surroundings; temperature of the system decreases.
- **Exothermic Process/Reaction:** Heat is released to surroundings; temperature of the system increases.

🧪 Endothermic vs. Exothermic — Comparison

Feature	Endothermic	Exothermic
Heat exchange	Absorbed from surroundings	Released to surroundings
Temperature effect	System cools down	System heats up
Process example	Melting of ice; dissolving KNO_3 in water	Freezing of water; dissolving $NaOH$ in water
Reaction example	$CaCO_3 \xrightarrow{\Delta} CaO +$	$CaO + H_2O \rightarrow Ca(OH$

Chemical Reaction Examples:

$$CaCO_3(s) + \text{Heat} \rightarrow CaO(s) + CO_2(g) \quad \text{(Endothermic Reaction)}$$

$$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(aq) + \text{Heat} \quad \text{(Exothermic Reaction)}$$

⚠️ **Safety Note:** While preparing dilute H_2SO_4, always add acid to water slowly (not water to acid). Adding water to concentrated H_2SO_4 releases large heat instantaneously, causing the water to evaporate explosively — very dangerous!

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7. Rate of a Chemical Reaction

Some reactions are fast (burning of gas, precipitation); others are slow (rusting of iron, erosion of rocks).

The same reaction can occur at different rates when conditions change.

Example: Milk sets into curd slowly in winter and faster in summer.

📈 Factors Affecting Rate of Reaction

a) 🔬 Nature of Reactants

- More reactive substances react at a faster rate.
- **Conclusion:** Al reacts faster with dilute HCl than Zn because Al is more reactive than Zn.

$$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2 \quad \text{(faster)}$$

$$Zn + 2HCl \rightarrow ZnCl_2 + H_2 \quad \text{(slower)}$$

b) 📦 Size of Reactant Particles

- Smaller particles $\rightarrow$ larger surface area $\rightarrow$ higher rate of reaction.
- **Conclusion:** Powder of Shahabad tile reacts faster with dilute HCl than its pieces, because powder has greater surface area exposed to acid.

Particle Size	Surface Area	Rate of Reaction
Large pieces	Small	Slow
Fine powder	Large	Fast

c) 🧪 Concentration of Reactants

- Higher concentration $\rightarrow$ faster reaction.
- **Conclusion:** Concentrated HCl reacts faster with $CaCO_3$ than dilute HCl.

$$CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2 \uparrow$$

- **Rate $\propto$ Concentration of reactants**

d) 🌡️ Temperature

- Higher temperature $\rightarrow$ faster reaction rate.
- **Conclusion:** Limestone ($CaCO_3$) does not decompose at room temperature; it requires heating. Rate of reaction increases with increasing temperature.

e) ⚡ Catalyst

- A substance whose presence increases the rate of a chemical reaction without itself undergoing any chemical change is called a **Catalyst**.
- **Example:** MnO_2 (manganese dioxide) acts as a catalyst in the decomposition of $KClO_3$:

$$2KClO_3 \xrightarrow[\Delta]{MnO_2} 2KCl + 3O_2 \quad \cdots (19)$$

- MnO_2 also speeds up the decomposition of H_2O_2:

$$2H_2O_2 \xrightarrow{MnO_2} 2H_2O + O_2 \uparrow$$

- **Key point:** MnO_2 undergoes no chemical change itself; it only increases the rate.

💡 In living organisms, enzymes act as biological catalysts to increase the rate of physiological reactions.

📊 **Summary: Factors Affecting Rate of Reaction**

Factor	Effect on Rate	Reason
Higher reactivity of reactants	Increases	More tendency to react
Smaller particle size	Increases	More surface area available
Higher concentration	Increases	More particles in contact
Higher temperature	Increases	Particles have more kinetic energy
Presence of catalyst	Increases	Provides alternative reaction pathway

8. Oxidation and Reduction (Redox Reactions)

🔴 Oxidation

A chemical reaction in which a reactant combines with oxygen OR loses hydrogen is called an Oxidation Reaction.

- In terms of electrons: **Oxidation = Loss of electrons.**

Examples:

$$2Mg + O_2 \rightarrow 2MgO \quad \cdots(20) \quad (\text{Mg gains oxygen — oxidised})$$

$$C + O_2 \rightarrow CO_2 \quad \cdots(21) \quad (\text{C gains oxygen — oxidised})$$

$$MgH_2 \rightarrow Mg + H_2 \uparrow \quad \cdots(22) \quad (\text{Mg loses hydrogen — oxidised})$$

$$CH_3 - CH_3 \rightarrow CH_2 = CH_2 + H_2 \uparrow \quad \cdots(23) \quad (\text{Ethane loses hydrogen — oxidised})$$

Oxidising Agents (Oxidants):

Chemical substances that bring about oxidation by making oxygen available are called oxidising agents or oxidants.

💡 **Nascent oxygen** $[O]$ is oxygen in a highly reactive state, just before the formation of the O_2 molecule.

Oxidation as electron loss — Example:

$$2KMnO_4 + 10FeSO_4 + 8H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 5Fe_2(SO_4)_3 + 8H_2O \quad \cdots (25)$$

Net ionic reaction:

$$Fe^{2+} \rightarrow Fe^{3+} + e^- \quad (\text{Ferrous ion} \rightarrow \text{Ferric ion})$$

- Fe^{2+} loses one electron $\rightarrow$ positive charge increases $\rightarrow$ Oxidation

Oxidant	Reaction	Nascent Oxygen Released
O_3 (Ozone)	$O_3 \rightarrow O_2 + [O]$	$[O]$
H_2O_2 (Hydrogen peroxide)	$H_2O_2 \rightarrow H_2O + [O]$	$[O]$
$K_2Cr_2O_7/H_2SO_4$	$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$	$3[O]$
$KMnO_4/H_2SO_4$	$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$	$5[O]$

🔵 Reduction

> A chemical reaction in which a reactant gains hydrogen OR loses oxygen is called a Reduction Reaction.

- In terms of electrons: **Reduction = Gain of electrons.**

Example:

$$CuO(s) + H_2(g) \rightarrow Cu(s) + H_2O(l) \quad \cdots (26)$$

- CuC loses oxygen $\rightarrow$ CuC is reduced (Copper oxide undergoes reduction)
- H_2 gains oxygen $\rightarrow$ H_2 is oxidised (Hydrogen undergoes oxidation)
- H_2 is the reductant (reducing agent) here.

💡 **Reduction = Gain of electrons.** When positive charge on an ion decreases or negative charge increases, it is reduction.

⚡ Redox Reactions

> Oxidation and Reduction always occur simultaneously. A single term "Redox Reaction" is used.

$$\text{Redox Reaction} = \text{Reduction} + \text{Oxidation}$$

Term	Definition	Process
Oxidation	Gain of oxygen / Loss of hydrogen / Loss of electrons	Charge increases
Reduction	Loss of oxygen / Gain of hydrogen / Gain of electrons	Charge decreases
Oxidant	Substance that causes oxidation (itself gets reduced)	Gains electrons
Reductant	Substance that causes reduction (itself gets oxidised)	Loses electrons

Oxidation State Progression of Iron:

$$Fe \rightarrow FeO \rightarrow Fe_2O_3$$

(Increasing oxidation state — progressive oxidation)

Example of Redox:

$$CuO + H_2 \rightarrow Cu + H_2O \quad \cdots (26)$$

Species	Change	Type
CuC	Loses $O \rightarrow$	Reduced
H_2	Gains $O \rightarrow$	Oxidised

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9. Corrosion 🦠

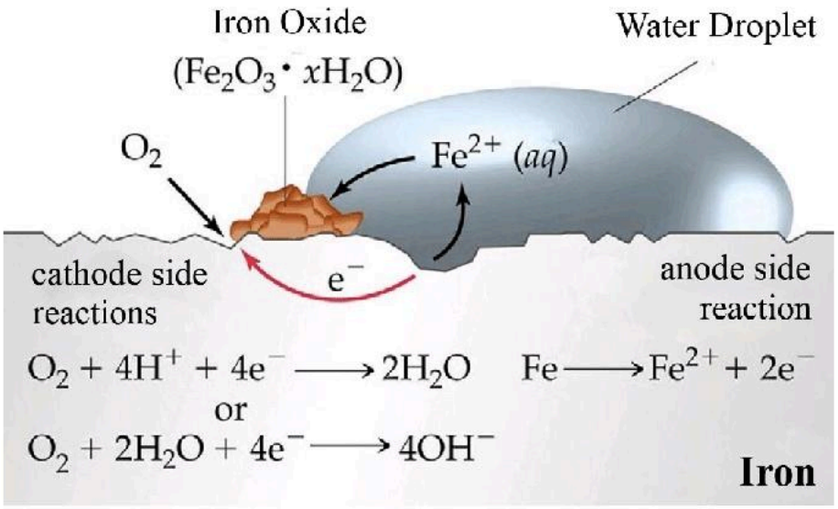
What is Corrosion?

> Due to various components of the atmosphere (oxygen, moisture, salts), oxidation of metals takes place, causing their degradation. This is called Corrosion.

- Rusting of iron is the most common example of corrosion.

- **Rust formula:** $Fe_2O_3 \cdot xH_2O$ (Hydrated iron oxide — reddish coloured layer)

⚙️ Mechanism of Rusting (Electrochemical Process)



Rust does NOT form by a simple reaction of O_2 with iron. It is an electrochemical reaction. Different regions of the iron surface become anode and cathode.

At the Anode (Oxidation):

$$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$$

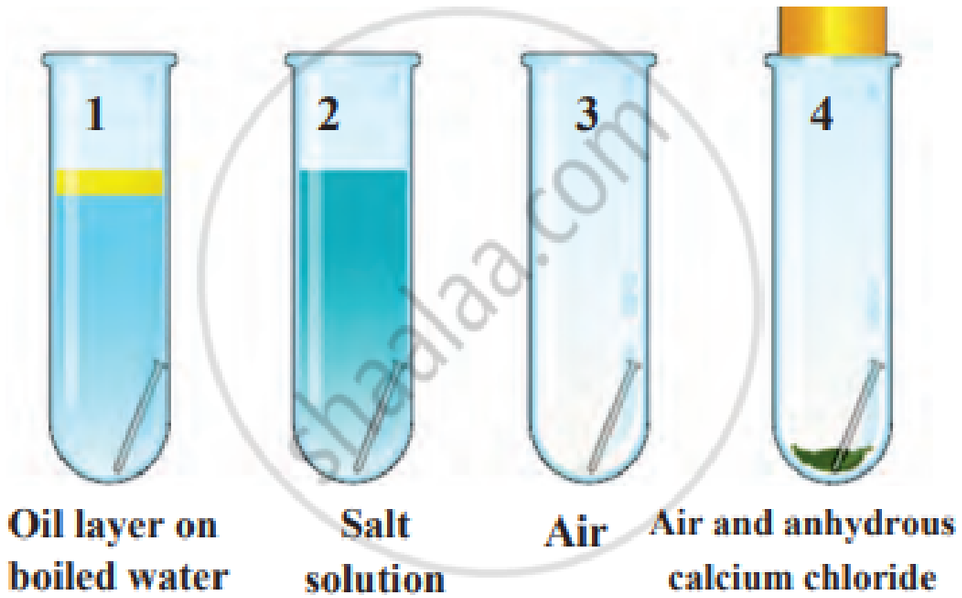
At the Cathode (Reduction):

$$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$$

Migration and further oxidation:

- Fe^{2+} ions migrate from the anode region and get further oxidised to Fe^{3+}.
- Fe^{3+} ions react with water to form rust:

$$2Fe^{3+}(aq) + 4H_2O(l) \rightarrow Fe_2O_3 \cdot H_2O(s) + 6H^+(aq) \quad \cdots (29)$$



🔬 Conditions Required for Rusting

Condition	Result
Only water (no air)	No rusting
Only air (no water)	No rusting
Both water AND air	Rusting occurs
Salt water + air	Rusting occurs rapidly
Dry air (anhydrous $CaCl_2$)	No rusting

Conclusion: Both water and air are necessary for rusting. Salt accelerates the process.

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10. Rancidity 🧈

What is Rancidity?

> When oils, ghee, or fried food is left for a long time, it undergoes air oxidation, developing a foul odour and changed taste. This is called Rancidity.

Prevention of Rancidity

Method	How it Works
Antioxidants	Added to food to slow down oxidation of oils/ghee
Air-tight containers	Prevents contact with atmospheric oxygen
Refrigeration	Low temperature slows oxidation reactions

🎯Formulating Chemical Compounds🎯

PART 1: Categorization of Radicals

1. Monovalent Radicals (Valency = 1)

These radicals need to give, take, or share exactly one electron to become stable.

Basic Radicals (+1)	Symbol	Acidic Radicals (-1)	Symbol
Hydrogen	H^+	Chloride	Cl^-
Sodium	Na^+	Bromide	Br^-
Potassium	K^+	Iodide	I^-
Silver	Ag^+	Fluoride	F^-
Ammonium	NH_4^+	Hydroxide	OH^-
Cuprous (Copper I)	Cu^+	Nitrate	NO_3^-
		Nitrite	NO_2^-
		Bicarbonate (Hydrogen Carbonate)	HCO_3^-
		Bisulphate (Hydrogen Sulphate)	HSO_4^-

2. Bivalent Radicals (Valency = 2)

These radicals have a combining capacity of two.

Basic Radicals (+2)	Symbol	Acidic Radicals (-2)	Symbol
Magnesium	Mg^{2+}	Oxide	O^{2-}
Calcium	Ca^{2+}	Sulphide	S^{2-}
Barium	Ba^{2+}	Sulphate	SO_4^{2-}
Zinc	Zn^{2+}	Sulphite	SO_3^{2-}
Cupric (Copper II)	Cu^{2+}	Carbonate	CO_3^{2-}
Ferrous (Iron II)	Fe^{2+}	Zincate	ZnO_2^{2-}
Lead	Pb^{2+}	Chromate	CrO_4^{2-}

3. Trivalent Radicals (Valency = 3)

These radicals have a combining capacity of three.

Basic Radicals (+3)	Symbol	Acidic Radicals (-3)	Symbol
Aluminium	Al^{3+}	Nitride	N^{3-}
Ferric (Iron III)	Fe^{3+}	Phosphide	P^{3-}
Chromium	Cr^{3+}	Phosphate	PO_4^{3-}
Gold (Auric)	Au^{3+}	Phosphite	PO_3^{3-}

4. Tetravalent Radicals (Valency = 4)

These radicals have a combining capacity of four.

Basic Radicals (+4)	Symbol	Acidic Radicals (-4)	Symbol
Stannic (Tin IV)	Sn^{4+}	Carbide	C^{4-}
Plumbic (Lead IV)	Pb^{4+}		
Silicon	Si^{4+}		

PART 2: Step-by-Step Method (The Criss-Cross Method)

Step 1: Write the Symbols

Always write the symbol of the **Basic Radical (Metal/Cation)** on the left and the **Acidic Radical (Non-metal/Anion)** on the right.

(Exception: In acetates like CH_3COONa *, the acidic radical is written first, but for Class 10 general inorganic salts, follow the standard rule).*

Step 2: Write the Valency

Write the numerical value of the valency directly below each respective symbol. Ignore the positive (+) or negative (-) signs.

Step 3: Simplify the Ratio (If possible)

If the valencies have a common factor, divide them by that factor to get a simple whole-number ratio. (e.g., if you have 2 and 2, simplify it to 1 and 1. If you have 2 and 4, simplify to 1 and 2).

Step 4: Criss-Cross!

"Criss-cross" the simplified valency numbers. The valency of the basic radical becomes the subscript of the acidic radical, and vice versa. Do not write the number '1' as a subscript; it is understood.

Step 5: Use Brackets for Polyatomic Ions

If an acidic or basic radical is made of *more than one atom* (a polyatomic ion like OH^-, SO_4^{2-}, NH_4^+) AND it receives a subscript greater than 1, you **must** enclose the entire polyatomic ion in brackets before writing the subscript outside.

PART 3: Practice Examples to do in Class

Example 1: Simple Binary Compound (No brackets needed)

Compound: Sodium Chloride

- **Step 1 (Symbols):** $Na \quad Cl$
- **Step 2 (Valency):** $1 \quad 1$
- **Step 3 (Simplify):** Already $1:1$
- **Step 4 (Criss-Cross):** Na_1Cl_1
- **Final Formula:** $NaCl$

Example 2: Polyatomic Ion (Brackets required)

Compound: Magnesium Hydroxide

- **Step 1 (Symbols):** $Mg \quad OH$
- **Step 2 (Valency):** $2 \quad 1$
- **Step 3 (Simplify):** Already simplest ratio ($2:1$)
- **Step 4 (Criss-Cross):** Mg_1 and OH_2
- **Step 5 (Brackets):** Because OH is polyatomic and gets a 2, use brackets.
- **Final Formula:** $Mg(OH)_2$

Example 3: Simplifying Valencies

Compound: Calcium Oxide

- **Step 1 (Symbols):** $Ca \quad O$
- **Step 2 (Valency):** $2 \quad 2$
- **Step 3 (Simplify):** Divide both by 2. The ratio becomes $1:1$.
- **Step 4 (Criss-Cross):** Ca_1O_1
- **Final Formula:** CaO

Example 4: Variable Valency & Two Polyatomic Ions

Compound: Ammonium Sulphate

- **Step 1 (Symbols):** $NH_4 \quad SO_4$
- **Step 2 (Valency):** $1 \quad 2$
- **Step 3 (Simplify):** Simplest ratio ($1:2$)
- **Step 4 (Criss-Cross):** NH_4 gets 2, SO_4 gets 1.
- **Step 5 (Brackets):** NH_4 is polyatomic and gets a 2, so bracket it. SO_4 only gets a 1, so no brackets needed for it.
- **Final Formula:** $(NH_4)_2SO_4$

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